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QUANTUM COMPUTING (BAI703D)

CCA Course Project Report

Quantum versus Classical Kernel Methods:

A Three-Way Comparison of RBF, Polynomial and Quantum Kernels

Breast Cancer Wisconsin and Iris (Versicolor vs. Virginica) Datasets

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1. Abstract

Support vector machines (SVMs) depend on a kernel function to measure similarity between data points, and this function can be replaced by one computed from a quantum circuit. This project compares three kernels for binary classification: a classical radial basis function (RBF) kernel, a classical polynomial kernel, and a quantum kernel built from a ZZ feature map and evaluated with Qiskit's `FidelityQuantumKernel` on the local Aer simulator. Two datasets are used: Breast Cancer Wisconsin (malignant versus benign) and the versicolor-versus-virginica subset of the Iris dataset. All three kernels operate on the same four features (one qubit per feature, obtained through standardisation and PCA for Breast Cancer), the same fixed stratified 70/30 split, and a common `SVC(kernel="precomputed")` evaluation path based on explicit Gram matrices. Performance is measured by test accuracy, five-fold cross-validation (CV) mean and standard deviation, and kernel-target alignment.

On Breast Cancer, the RBF kernel reached a test accuracy of 0.9833 and a CV accuracy of 0.9429 ± 0.0286 , whereas the quantum kernel reached 0.6833 and 0.6071 ± 0.0639 . On Iris, the RBF kernel reached a CV accuracy of 0.9714 ± 0.0350 , against 0.8286 ± 0.1069 for the quantum kernel. The quantum kernel did not satisfy the pre-specified advantage criterion on either dataset. The project therefore reports a null result: no quantum advantage was observed for this feature map, these datasets, and this local Aer simulation setting (no noise model).

Keywords: quantum kernel, support vector machine, ZZ feature map, `FidelityQuantumKernel`, kernel-target alignment, Qiskit, Aer.

2. Introduction

A kernel SVM classifies data through a kernel function $k(x, x')$ that returns a similarity score for every pair of samples. When the kernel is evaluated on all pairs of training points it forms a Gram matrix, which is the only information about the data that the SVM optimisation requires [4]. Quantum kernel methods use a quantum circuit as the feature map: each classical sample is encoded into a quantum state, and the overlap between the states of two samples is used as their kernel value [1], [2]. Whether such a kernel offers a measurable benefit on practical classical data is an empirical question, and it is the question addressed by this project.

The project, carried out for the course Quantum Computing (BAI703D), implements a controlled three-way comparison between an RBF-kernel SVM, a polynomial-kernel SVM, and an SVM using a simulated quantum kernel. The quantum kernel is built with Qiskit [6] using a ZZ feature map, the `FidelityQuantumKernel` class of Qiskit Machine Learning, and the local Aer simulator; no real quantum hardware is used. The implementation is provided as two Jupyter notebooks, one per dataset, which share identical code and differ only in the dataset selected.

Two datasets with different characteristics are used. The Breast Cancer Wisconsin dataset has 30 features, which would require a 30-qubit circuit if one qubit were assigned per feature, so it is reduced to four principal components. The Iris dataset has four features and is restricted to the versicolor and virginica classes, which overlap and therefore form a more demanding binary problem than any pair involving setosa. The rest of this report describes the objectives, the methodology and datasets, the experimental setup and evaluation metrics, the results for each dataset, and a discussion of whether the quantum kernel provides an advantage on the evidence of the project.

3. Objectives

The objectives of the project are as follows:

1. To implement classical RBF-kernel and polynomial-kernel SVMs with hyperparameters selected by cross-validated grid search.
2. To implement a quantum-kernel SVM using a ZZ feature map and `FidelityQuantumKernel`, with the quantum kernel (Gram) matrix computed on the local Aer simulator.
3. To compare the three kernels fairly by using identical features, one fixed train/test split, and one shared evaluation pathway based on precomputed Gram matrices.
4. To evaluate each kernel using test accuracy, five-fold cross-validation mean and standard deviation, and kernel-target alignment, and to visualise the resulting kernel matrices.
5. To determine, using a criterion specified in advance, whether the quantum kernel shows a measurable advantage over the best classical baseline on the Breast Cancer and Iris datasets.

4. Methodology

The methodology follows the pipeline in Figure 1. For each dataset the data are sub-sampled where required, split once into training and test sets, reduced to four features where required, and scaled to the interval $[0, \pi]$. Each of the three kernels is then converted into an explicit Gram matrix: an $N_{\text{train}} \times N_{\text{train}}$ matrix over the training samples and an $N_{\text{test}} \times N_{\text{train}}$ matrix between test and training samples. Every Gram matrix is evaluated through the same `SVC(kernel="precomputed")` pathway, so that fitting, test scoring, k-fold cross-validation (performed on sub-matrices of the already-computed training Gram matrix, without recomputing any kernel per fold), and kernel-target alignment all use identical machinery for the classical and quantum kernels.

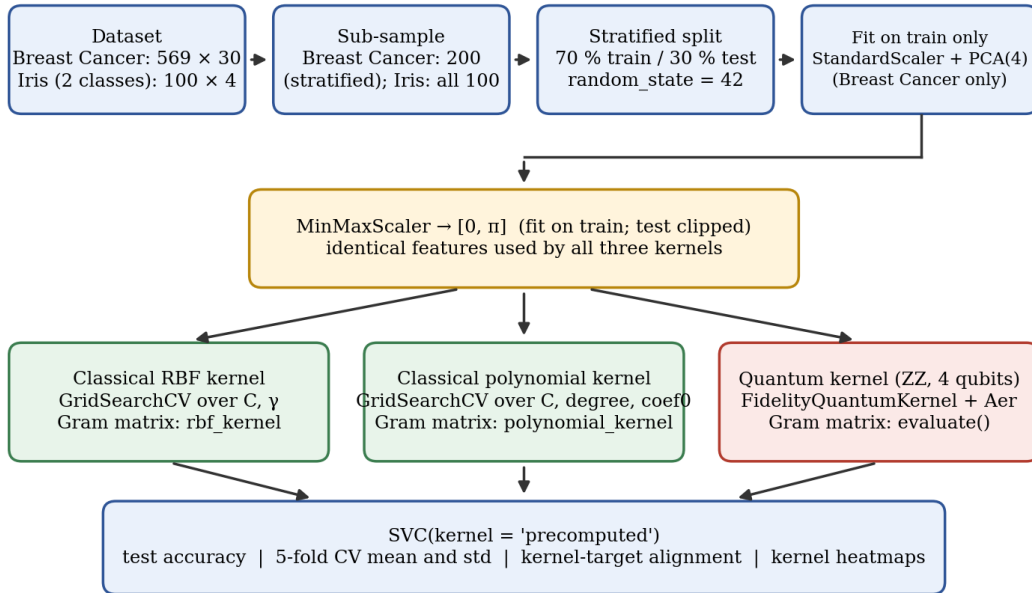


Figure 1: Methodology pipeline used for both datasets. All three kernels receive identical scaled features and are evaluated through the same precomputed-kernel SVM pathway.

4.1 Classical RBF Kernel

The RBF kernel measures similarity as a decaying function of the squared Euclidean distance between two samples:

$$k_{\text{RBF}}(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (1)$$

The hyperparameters C and γ were selected with `GridSearchCV` (five-fold, native scikit-learn `SVC(kernel="rbf")`) over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.01, 0.1, 1, \text{"scale"}\}$. When “scale” was selected, the heuristic was reproduced explicitly as $\gamma = 1 / (n_{\text{features}} \cdot \text{Var}(X_{\text{train}}))$, so that the Gram matrix rebuilt with `rbf_kernel` is identical to the kernel

scored by the grid search. The resulting training Gram matrix K_{train} and the test-versus-train matrix K_{test} were passed to the shared precomputed-kernel pathway using the selected C .

4.2 Classical Polynomial Kernel

The polynomial kernel is defined as

$$k_{\text{poly}}(x, x') = (\gamma \langle x, x' \rangle + r)^d \quad (2)$$

where d is the degree, r is the independent term (coef0), and γ is the scale factor. A five-fold GridSearchCV over $C \in \{0.1, 1, 10\}$, $d \in \{2, 3, 4\}$, and $r \in \{0, 1\}$ selected the polynomial hyperparameters. The scikit-learn `SVC(kernel="poly")` class defaults to $\gamma = \text{"scale"}$, whereas the `polynomial_kernel()` function defaults to $\gamma = 1 / n_{\text{features}}$. To ensure that the rebuilt Gram matrix equals the kernel that the grid search actually scored, the project passes $\gamma = 1 / (n_{\text{features}} \cdot \text{Var}(X_{\text{train}}))$ explicitly. The Gram matrices were then evaluated through the shared precomputed-kernel pathway.

4.3 Quantum Kernel

ZZ feature map. The quantum kernel uses Qiskit’s ZZ feature map on four qubits (one qubit per input feature), configured with two repetitions (reps=2) and linear entanglement. The circuit is created with the `zz_feature_map()` function, which returns a circuit already composed of standard gates. The older `ZZFeatureMap` class is deprecated in current Qiskit, and, as noted in the notebook, its opaque block must otherwise be decomposed before Aer can execute it; the notebook falls back to that decomposed form only on older Qiskit versions. The state prepared for an input x is

$$|\varphi(x)\rangle = (U_Z(x) H^{\otimes 4})^2 |0\rangle^{\otimes 4} \quad (3)$$

Each repetition applies a Hadamard gate to every qubit, a single-qubit phase gate $P(\theta_i)$ on every qubit, and, for each neighbouring qubit pair, a CX–P–CX block that implements the two-qubit phase. As shown in the project circuit (Figure 2), the phase angles are

$$\theta_i = 2x_i, \quad \theta_{i,i+1} = 2(\pi - x_i)(\pi - x_{i+1}) \quad (4)$$

with the pairs (0, 1), (1, 2), and (2, 3) entangled. The resulting circuit has a depth of 19 and contains 8 Hadamard (H), 14 phase (P), and 12 controlled-NOT (CX) gates, which corresponds to two repetitions of four H gates, four single-qubit P gates plus three pair P gates, and three CX–P–CX blocks. The circuit is independent of the dataset; only the four input values differ.

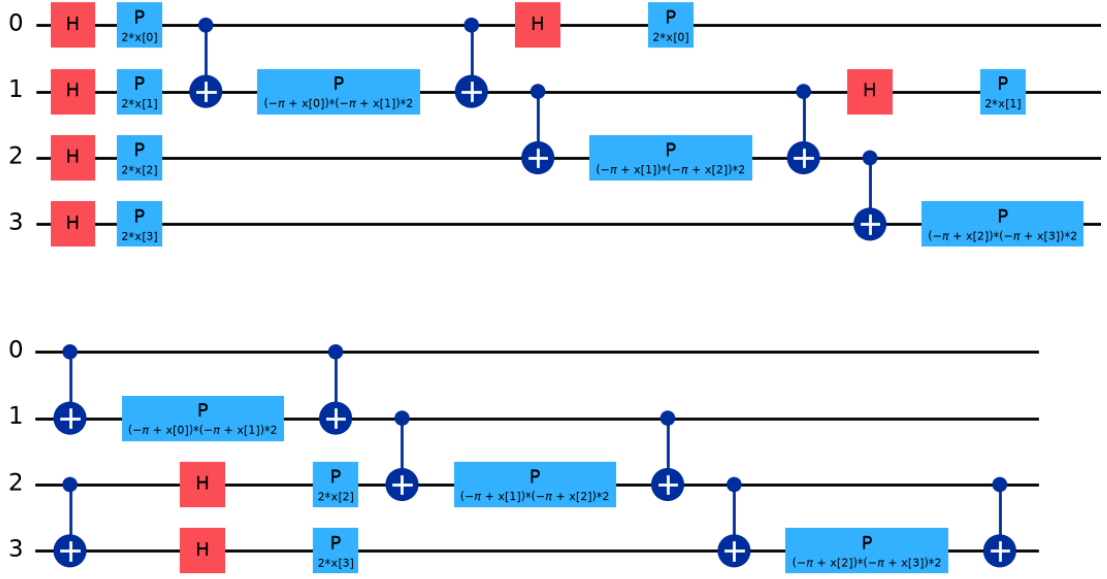


Figure 2: ZZ feature map circuit (4 qubits, reps=2, linear entanglement) as drawn by the project notebook. The single circuit is shown in two consecutive rows, read from top to bottom; the $x[i]$ labels are the four scaled input features.

Fidelity-based kernel and Gram matrix. The quantum kernel value for two samples is the fidelity between their encoded states:

$$k_Q(x, x') = |\langle \varphi(x) | \varphi(x') \rangle|^2 \quad (5)$$

In the project this quantity is estimated with the compute–uncompute method, through the `ComputeUncompute` fidelity class of Qiskit Machine Learning, which uses a sampler to run the circuit and read out the all-zero outcome probability. The sampler is the Qiskit Aer `SamplerV2` primitive, created with a default of 1024 shots and run on the local Aer simulator. The `FidelityQuantumKernel` object, constructed from the feature map and this fidelity object, produces the quantum kernel (Gram) matrix. The training Gram matrix is obtained with `evaluate(x_vec=X_train)`, and the test-versus-train kernel matrix with `evaluate(x_vec=X_test, y_vec=X_train)`. The diagonal of the training Gram matrix is equal to one, since each sample has unit fidelity with itself.

Classification. The quantum Gram matrices are passed to the same `SVC(kernel="precomputed")` pathway as the classical kernels. The regularisation parameter was fixed at $C = 1.0$ and was not tuned. The notebook documents this as a fair-comparison choice that avoids tuning against the test set, and the consequence of this choice is discussed in Section 10.

5. Datasets and Preprocessing

Both datasets are loaded from `sklearn.datasets`, so no external download is required. Table 1 summarises the data used in each experiment. For both datasets, all transformations (standardisation, PCA, and scaling to $[0, \pi]$) are fitted on the training split only and then applied to the test split, which prevents information from the test set from leaking into the transformations.

Table 1: Summary of the two datasets and the resulting train/test configuration.

Property	Breast Cancer Wisconsin	Iris (versicolor vs. virginica)
Source	<code>load_breast_cancer</code> (Diagnostic)	<code>load_iris</code> , classes 1 and 2
Samples in full dataset	569	150 (100 after class selection)
Original features	30	4
Class labels	0 = malignant (212); 1 = benign (357)	0 = versicolor (50); 1 = virginica (50)
Samples used	200 (class-balanced sub-sample)	100 (no sub-sampling)
Dimensionality reduction	StandardScaler + PCA to 4 components	None (4 features)
Final feature range	$[0, \pi]$	$[0, \pi]$
Training set (class 0 / class 1)	140 (52 / 88)	70 (35 / 35)
Test set (class 0 / class 1)	60 (23 / 37)	30 (15 / 15)
Qubits in the feature map	4	4

5.1 Breast Cancer Wisconsin Dataset

The Breast Cancer Wisconsin (Diagnostic) dataset [7] contains 569 samples described by 30 numerical features, with 212 malignant cases (label 0) and 357 benign cases (label 1). Because one qubit is used per feature, using all 30 features would require a 30-qubit circuit, which is far beyond what simulates quickly and beyond the qubit count that the comparison is intended to exercise. The data were therefore reduced to four dimensions. The preprocessing steps, in order, were:

1. A class-balanced (stratified) sub-sample of 200 of the 569 points was drawn (`SUBSAMPLE_SIZE = 200`), preserving the class proportions (37.5 % malignant, 62.5 % benign). The notebook documents this as a runtime decision, since quantum kernel evaluation requires $O(N^2)$ fidelity circuits.

2. The sub-sample was split once into 140 training and 60 test points (TEST_FRACTION = 0.30, stratified, random_state = 42). The same split is used by all three kernels.
3. StandardScaler was fitted on the training data, followed by PCA with four components, also fitted on the standardised training data.
4. MinMaxScaler with feature_range=(0, π) was fitted on the PCA-reduced training data; test points were transformed with the same scaler and clipped to [0, π]. The same scaled features were reused for all three kernels.
5. Labels were kept as {0, 1} for the SVM and converted to {-1, +1} for the kernel-target alignment calculation.

The four principal components retain 80.86 % of the total variance of the standardised training data (Table 2). All three kernels receive these same four components, so any information lost in the reduction affects the classical and quantum kernels equally.

Table 2: Explained variance of the four PCA components for Breast Cancer (fitted on the 140 training points).

Component	Explained variance ratio
PC1	0.4323
PC2	0.2150
PC3	0.0920
PC4	0.0692
Total	0.8086

5.2 Iris Dataset – Versicolor vs. Virginica

Fisher’s Iris dataset [8] contains 150 samples of three classes with four features each. The kernel-target alignment metric used in this project is defined for binary labels, so a two-class subset was required. The versicolor and virginica classes (class identifiers 1 and 2) were retained, which gives 100 samples, 50 per class. This pair was chosen because it is the genuinely overlapping pair: setosa is linearly separable from both other classes, so any pair containing setosa would give approximately 100 % accuracy for every kernel and make the comparison uninformative, as stated in the notebook.

The two retained classes were relabelled so that versicolor is 0 and virginica is 1. Because Iris already has four features, no PCA was applied and the four original features map directly to the four qubits. The 100 samples were used in full (the sub-sampling setting of

200 does not take effect), and were split once into 70 training points (35 per class) and 30 test points (15 per class) with a stratified split and `random_state = 42`. As for Breast Cancer, a `MinMaxScaler` mapping features to $[0, \pi]$ was fitted on the training data, the test data were clipped to the same interval, and the same scaled features were used by all three kernels.

6. Experimental Setup

All experiments were executed locally in a Python 3.12.10 virtual environment. Table 3 lists the software versions recorded by the notebooks and the configuration of the quantum kernel simulation. The notebook first attempted to initialise the Aer simulator on a GPU; this was unavailable (a `RuntimeError` was raised), and the notebook then used the CPU device automatically. The notebooks use only the local Aer simulator; they do not authenticate with IBM Quantum and do not submit jobs to real quantum processors.

Table 3: Software environment and quantum kernel simulation configuration.

Item	Configuration
Python	3.12.10 (virtual environment)
Qiskit / Qiskit Aer	2.5.2 / 0.17.2
Qiskit Machine Learning	0.9.1
scikit-learn / NumPy	1.9.1 / 2.5.3
Simulator	Qiskit Aer SamplerV2 primitive (local simulator), method “automatic”
Device used	CPU (GPU initialisation attempted first and was unavailable)
Sampler shots	Default of 1024
Feature map	<code>zz_feature_map(feature_dimension=4, reps=2, entanglement="linear")</code> ; 4 qubits
Fidelity / kernel	<code>ComputeUncompute / FidelityQuantumKernel</code>
Data split	30 % test, stratified, <code>random_state = 42</code>
Cross-validation	<code>StratifiedKFold</code> , 5 splits, shuffled, <code>random_state = 42</code>
Quantum hardware	None used

Classical hyperparameter selection. The search grids and the selected values are given in Table 4. The grid search was run on the training split with five-fold cross-validation. The quantum kernel was not tuned: its feature-map configuration and $C = 1.0$ were fixed in advance.

Table 4: Hyperparameter search grids and selected values.

Kernel	Search grid	Breast Cancer (selected)	Iris (selected)
RBF	$C \in \{0.1, 1, 10, 100\}; \gamma \in \{0.01, 0.1, 1, \text{"scale"}\}$	$C = 1; \gamma = 0.1$	$C = 0.1; \gamma = \text{"scale"} (\approx 0.4628)$
Polynomial	$C \in \{0.1, 1, 10\}; d \in \{2, 3, 4\}; r \in \{0, 1\}$	$C = 0.1; d = 4; r = 0$	$C = 0.1; d = 2; r = 0$
Quantum	None (C fixed; ZZ map with reps=2, linear)	$C = 1.0$	$C = 1.0$

Cost of the quantum kernel. For N_{train} training and N_{test} test points, the quantum kernel requires $N_{\text{train}}(N_{\text{train}} - 1)/2$ fidelity circuits for the training Gram matrix and $N_{\text{test}} \cdot N_{\text{train}}$ circuits for the test matrix. Table 5 gives the matrix sizes, the number of circuits, and the measured computation time on the CPU for each dataset.

Table 5: Quantum kernel evaluation on the local Aer simulator (CPU).

Quantity	Breast Cancer	Iris
Qubits	4	4
Training Gram matrix	140×140	70×70
Test-versus-train matrix	60×140	30×70
Fidelity circuits evaluated	18,130 (9,730 + 8,400)	4,515 (2,415 + 2,100)
Computation time	1009.8 s	116.3 s

7. Evaluation Metrics

Test accuracy. The SVM is fitted on the training Gram matrix and used to predict the held-out test points through the test-versus-train kernel matrix. Test accuracy is the fraction of test points classified correctly. Because there are only 60 (Breast Cancer) and 30 (Iris) test points, a single misclassification changes the accuracy by 1.67 and 3.33 percentage points respectively.

Cross-validation mean and standard deviation. Stratified five-fold cross-validation is performed on the training set only, using sub-matrices of the precomputed training Gram matrix. If a_f is the validation accuracy of fold f , the reported mean and standard deviation are

$$\mu = \frac{1}{5} \sum_{f=1}^5 a_f \quad (6)$$

$$\sigma = \sqrt{\frac{1}{5} \sum_{f=1}^5 (a_f - \mu)^2} \quad (7)$$

The standard deviation is the population form (division by the number of folds, as in NumPy's default), and it measures how much accuracy varies across folds.

Kernel-target alignment (KTA). Kernel-target alignment [3] measures how closely a kernel matrix K agrees with the ideal kernel yy^T built from the labels $y \in \{-1, +1\}^n$:

$$A(K, y) = \frac{\langle K, yy^T \rangle_F}{\|K\|_F \|yy^T\|_F} \quad (8)$$

where $\langle \cdot, \cdot \rangle_F$ and $\|\cdot\|_F$ are the Frobenius inner product and norm. In the project, KTA is computed on the 140×140 (Breast Cancer) or 70×70 (Iris) training Gram matrix of each kernel. A higher alignment indicates that the kernel geometry is more closely related to the class labels, but it does not by itself guarantee better generalisation.

Kernel matrix heatmaps. The training Gram matrices are visualised as heatmaps with rows and columns sorted by class label. A kernel that separates the classes well should show a visible 2×2 block structure, with high similarity within each class and lower similarity between classes. This is a qualitative diagnostic.

Advantage criterion. To avoid over-claiming on small samples, the notebook applies a conservative, pre-specified criterion. Let Δ_{acc} and Δ_{KTA} be the differences (quantum minus best classical baseline, where the baseline is the classical kernel with the higher CV mean accuracy) in CV mean accuracy and alignment, and let σ_Q and σ_C be the CV standard deviations of the quantum kernel and the baseline. The quantum kernel is declared to show a measurable advantage only if both of the following conditions hold:

$$\Delta_{acc} > \sqrt{\sigma_Q^2 + \sigma_C^2}, \quad \Delta_{KTA} > 0 \quad (9)$$

8. Results and Comparison

All values in this section are taken from the outputs stored in the project notebooks. For every dataset, all three kernels use the same features, the same train/test split, and the same evaluation functions. The results are from one run with one train/test split (`random_state = 42`).

8.1 Breast Cancer Results

Table 6: Results on Breast Cancer Wisconsin (140 training points, 60 test points, 4 qubits/features).

Kernel	N (train)	Test accuracy	CV mean accuracy	CV std	Kernel-target alignment
Classical RBF	140	0.9833	0.9429	0.0286	0.1449
Classical Polynomial	140	0.9667	0.9214	0.0416	0.0931
Quantum (local simulation)	140	0.6833	0.6071	0.0639	0.1009

The classical kernels perform strongly on the four PCA components. The RBF kernel classifies 59 of the 60 test points correctly (0.9833) and the polynomial kernel 58 of 60 (0.9667), with CV accuracies of 0.9429 ± 0.0286 and 0.9214 ± 0.0416 . The quantum kernel classifies 41 of 60 test points correctly (0.6833), and its CV accuracy is 0.6071 ± 0.0639 , which is the lowest of the three and has the largest fold-to-fold spread. For reference, the majority class (benign) makes up 88 of 140 training points (0.6286) and 37 of 60 test points (0.6167).

In terms of alignment, the RBF kernel has the highest value (0.1449). The quantum kernel (0.1009) is slightly above the polynomial kernel (0.0931), even though its accuracy is far lower. The kernel matrices are shown in Figure 3. The RBF matrix shows visible class-dependent block structure, with a high-similarity region for the benign class. The polynomial matrix is on a much larger, unnormalised scale (colour bar up to about 3×10^4), and any class structure is much less visible. The quantum matrix has a unit diagonal and low off-diagonal values with scattered higher entries, and no clear 2×2 block pattern is visible.

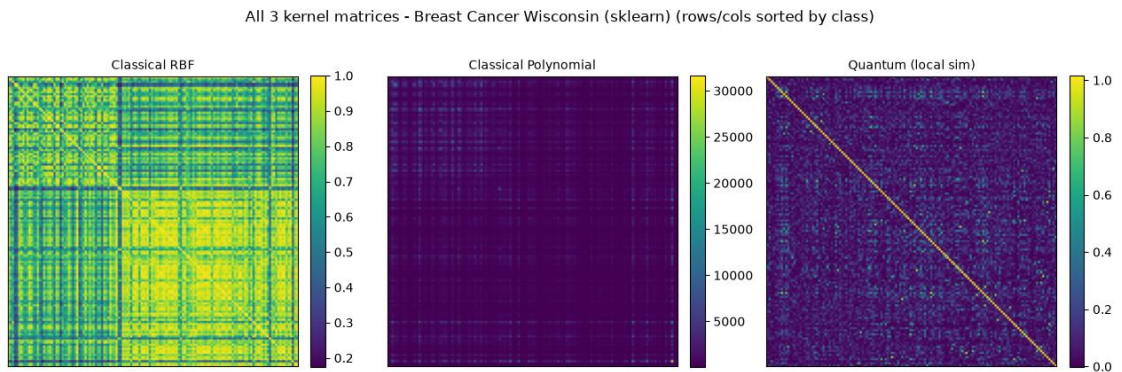


Figure 3: Training kernel matrices for Breast Cancer (rows and columns sorted by class): classical RBF (left), classical polynomial (centre), and quantum kernel from local simulation (right). Output of the project notebook.

8.2 Iris Results

Table 7: Results on Iris, versicolor vs. virginica (70 training points, 30 test points, 4 qubits/features).

Kernel	N (train)	Test accuracy	CV mean accuracy	CV std	Kernel-target alignment
Classical RBF	70	0.8667	0.9714	0.0350	0.3674
Classical Polynomial	70	0.8667	0.9429	0.0286	0.2191
Quantum (local simulation)	70	0.7333	0.8286	0.1069	0.1501

On Iris, the RBF and polynomial kernels obtain the same test accuracy (0.8667, i.e. 26 of 30 test points), while the quantum kernel obtains 0.7333 (22 of 30). Cross-validation on the training set gives a clearer ordering: RBF (0.9714 ± 0.0350), polynomial (0.9429 ± 0.0286), and quantum (0.8286 ± 0.1069). The quantum kernel again has the lowest mean and by far the largest standard deviation, roughly three to four times that of the classical kernels. The test accuracies of the classical kernels are lower than their CV accuracies, which reflects how coarse a 30-point test set is: each misclassification changes the test accuracy by 3.33 percentage points.

The alignment values follow the same ordering as the CV accuracy: RBF (0.3674), polynomial (0.2191), and quantum (0.1501). The kernel matrices are shown in Figure 4. The RBF matrix has a clearly visible 2×2 block structure for the two classes. The polynomial matrix (colour scale up to about 240) shows weaker structure, and the quantum matrix is again dominated by its unit diagonal, with mostly low off-diagonal values and no clear block pattern.

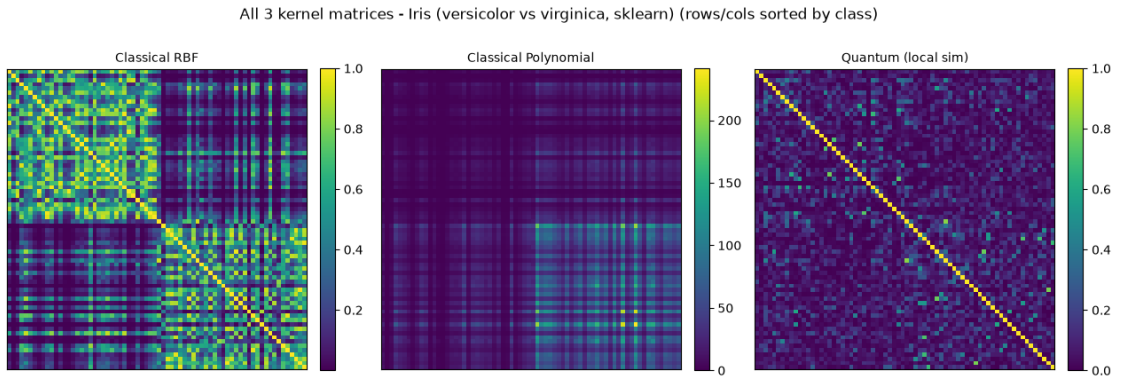


Figure 4: Training kernel matrices for Iris, versicolor vs. virginica (rows and columns sorted by class): classical RBF (left), classical polynomial (centre), and quantum kernel from local simulation (right). Output of the project notebook.

8.3 Overall Comparison

Table 8 applies the advantage criterion of Equation (9) to both datasets, and Figure 5 summarises all reported metrics. In both cases the best classical baseline is the RBF kernel.

Table 8: Quantum kernel versus the best classical baseline (RBF) under the pre-specified advantage criterion.

Dataset	Best classical CV mean	Quantum CV mean	Δ_{acc}	Combined std	Δ_{KTA}	Advantage shown?
Breast Cancer	0.9429	0.6071	-0.3357	0.0700	-0.0440	No
Iris	0.9714	0.8286	-0.1429	0.1125	-0.2174	No

Combined std = $\sqrt{(\sigma_Q^2 + \sigma_C^2)}$. Δ values are quantum minus best classical, as printed by the notebook.

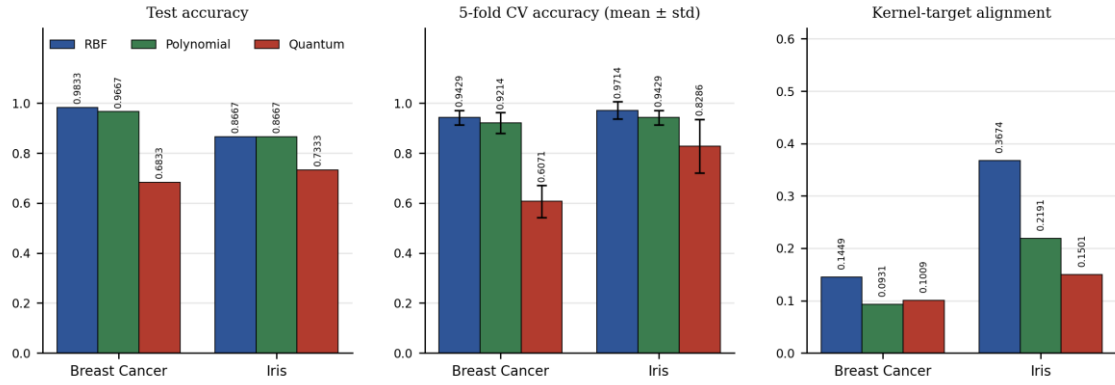


Figure 5: Test accuracy, five-fold CV accuracy (mean $\pm$ standard deviation), and kernel-target alignment for the three kernels on both datasets, plotted from the values in Tables 6 and 7.

The quantum kernel has the lowest test accuracy, the lowest CV mean accuracy, and the highest CV standard deviation on both datasets. It also has the lowest alignment on Iris, and on Breast Cancer its alignment is below that of RBF. On Breast Cancer the CV accuracy deficit relative to the RBF baseline is about 4.8 combined standard deviations ($0.3357 / 0.0700$), and on Iris it is about 1.3 ($0.1429 / 0.1125$). The criterion of Equation (9) is not satisfied on either dataset.

9. Discussion

Quantum advantage. Based strictly on the project results, the quantum kernel provides no advantage. It does not exceed the best classical kernel on any accuracy metric, on either dataset, and it fails the pre-specified criterion of Equation (9) in both cases. The deficit is large on Breast Cancer (CV accuracy 0.6071 versus 0.9429) and smaller, but still consistently negative, on Iris (0.8286 versus 0.9714). The notebook therefore reports a null result. The finding applies to the specific combination tested here: a four-qubit ZZ feature

map with two repetitions and linear entanglement, the preprocessing described in Section 5, an untuned C, and a local simulation without a noise model. It does not show that quantum kernels are inferior in general.

Accuracy and alignment. Kernel-target alignment does not predict accuracy reliably in these results. On Iris the alignment ordering (RBF > polynomial > quantum) matches the CV ordering. On Breast Cancer, however, the quantum kernel has a higher alignment (0.1009) than the polynomial kernel (0.0931) while achieving a far lower accuracy (0.6071 versus 0.9214 in CV). This agrees with the cautionary statement in the notebook that higher alignment does not by itself guarantee better generalisation. All alignment values in the project are small in absolute terms (at most 0.3674).

Kernel-matrix structure. The heatmaps are consistent with the accuracy results. The RBF kernel yields visible class-related block structure, most clearly on Iris. The quantum kernel matrices are dominated by the unit diagonal, with mostly low off-diagonal fidelities and no clear block pattern. A matrix of this form carries little similarity information between different samples, which is consistent with the weaker classification performance. The project did not run an experiment that varies the feature-map repetitions, entanglement, or data scaling, so the underlying cause of this structure is not isolated and is not asserted here.

Role of preprocessing. On Breast Cancer, all three kernels receive identical four-dimensional PCA features that retain 80.86 % of the variance. The classical kernels reach CV accuracies of 0.92 or higher on these same features, so the information lost in the PCA reduction cannot by itself explain the weak quantum result. As a point of reference, the quantum test accuracy (0.6833) is only moderately above the majority-class proportion of the test set (0.6167), and its CV accuracy (0.6071) is slightly below the training majority proportion (0.6286). The Iris classes are balanced (chance level 0.5), and the quantum CV accuracy (0.8286) is well above this level.

Variability and sample size. The quantum kernel has the largest CV standard deviation on both datasets (0.0639 on Breast Cancer; 0.1069 on Iris, against 0.0286–0.0416 and 0.0286–0.0350 for the classical kernels). The results come from one split of small datasets, so they should be read as indicative. This is especially true for the Iris test accuracies, where the classical kernels score 0.8667 on the test set but 0.9714 and 0.9429 in cross-validation, and the test set contains only 30 points.

Computational cost. The quantum Gram matrices required 18,130 fidelity circuits (1009.8 s) for Breast Cancer and 4,515 circuits (116.3 s) for Iris. The number of circuits increased by a factor of about 4.0 between the two datasets, and the computation time by about 8.7. The notebook identifies the quantum kernel evaluation as its slowest cell, while the classical Gram matrices are obtained directly from scikit-learn functions. Since the quantum kernel did not deliver higher accuracy, the project shows no benefit that offsets this additional cost.

Fairness of the comparison. The classical kernels were tuned by grid search whereas the quantum kernel used fixed settings ($C = 1.0$, one feature-map configuration). The comparison therefore does not give the quantum kernel the same tuning effort. The notebook documents this as a deliberate choice to avoid tuning against the test set; whether tuning C or the feature map would change the outcome was not tested.

10. Limitations

The conclusions of this report are subject to the following limitations of the project:

- **Small samples and a single split.** Breast Cancer uses a 200-point sub-sample of 569 (140 training, 60 test) and Iris uses 100 points (70 training, 30 test). Only one stratified split with `random_state = 42` was evaluated, so the variability across different splits or seeds is unknown. No statistical significance test was performed.
- **Cross-validation scope.** Cross-validation was performed within the training set only (140 and 70 points), and the standard deviation is based on five folds. Kernel-target alignment is a single value per kernel computed on the training Gram matrix, without an uncertainty estimate.
- **Dimensionality reduction.** For Breast Cancer, PCA to four components retains 80.86 % of the variance, so the experiment evaluates the kernels on a reduced representation rather than on all 30 features.
- **One quantum configuration.** Only one ZZ feature-map configuration (four qubits, `reps=2`, linear entanglement) was evaluated. Other repetitions, entanglement patterns, numbers of qubits, or encodings were not explored.
- **Unequal tuning.** The classical kernels were tuned by cross-validated grid search, whereas the quantum kernel used a fixed $C = 1.0$.

- **Simulation only.** The quantum kernel was run on the local Aer simulator on a CPU, with a default of 1024 shots, and no noise model is configured in the notebooks. No real quantum hardware was used, so the results say nothing about hardware performance.
- **Scalability.** Quantum Gram matrix evaluation requires $O(N^2)$ fidelity circuits, which restricted the sample sizes and made the Breast Cancer kernel computation take 1009.8 s even for four qubits and 200 points.
- **Scope of the heatmap analysis.** The kernel-matrix interpretation is visual and qualitative, and the project did not run experiments to isolate why the quantum matrices have the structure they do.

11. Conclusion and Learning Outcomes

This project compared classical RBF and polynomial SVM kernels with a four-qubit ZZ-feature-map quantum kernel evaluated through `FidelityQuantumKernel` on the local Aer simulator, using the Breast Cancer Wisconsin and Iris (versicolor versus virginica) datasets. All kernels were evaluated on identical features and an identical split through a common precomputed-kernel pathway, using test accuracy, five-fold CV mean and standard deviation, and kernel-target alignment. The RBF kernel was the strongest on both datasets (CV accuracy 0.9429 ± 0.0286 on Breast Cancer and 0.9714 ± 0.0350 on Iris), while the quantum kernel obtained 0.6071 ± 0.0639 and 0.8286 ± 0.1069 respectively. The quantum kernel did not meet the pre-specified advantage criterion on either dataset, so the project concludes that no quantum advantage was observed for this feature map and these datasets in simulation. This null result is reported as a valid outcome of the experiment, subject to the limitations in Section 10.

Learning outcomes. Through this project the following were learned:

- How to build a quantum kernel in Qiskit with a ZZ feature map and `FidelityQuantumKernel`, and how its Gram matrix is obtained from fidelity circuits executed on the Aer simulator.
- How to evaluate classical and quantum kernels through one shared `SVC(kernel="precomputed")` pathway, including cross-validation on sub-matrices of a precomputed Gram matrix.

- How to prepare data for angle-encoding feature maps: one qubit per feature, PCA for high-dimensional data, scaling to $[0, \pi]$, and fitting every transformation on the training set only to avoid leakage.
- How to use kernel-target alignment and kernel heatmaps alongside accuracy, and why alignment alone does not guarantee better generalisation.
- How the $O(N^2)$ circuit cost of quantum kernels limits sample size in simulation.
- How to report a null result honestly by applying a pre-specified advantage criterion rather than assuming a quantum benefit.

12. References

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